

MAXIME ROCHE

Department of Economics & Public Policy
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PERSONAL DATA

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EDUCATION

2022-2026 PhD Candidate in ECONOMICS & PUBLIC POLICY, **Imperial College London** – UK
Affiliation: Centre for Health Economics & Policy Innovation
Advisors: Prof. Franco SASSI, Dr. Laure DE PREUX
Expected graduation: Spring 2026

2021-2022 Awarded **Distinction** for MRes in ECONOMICS & PUBLIC POLICY
Imperial College London – UK

2017-2018 Master of Sc. in ECONOMICS, **University of Amsterdam** – Netherlands

2013-2017 Master of Sc. in CORPORATE FINANCE & BANKING, **EDHEC Business School**
Bachelor's degree in BUSINESS MANAGEMENT, **EDHEC Business School** – France

2011-2013 Preparatory classes in MATHEMATICS & PHYSICS for the entrance exams to top
French engineer & business schools, **Lycee Joffre (Montpellier)** – France

VISITING

SPRING 2025 Visiting researcher, hosted by DR. ARIEL ORTIZ BOBEA
Cornell University, Dyson School of Applied Economics & Management – NY, USA

SPRING 2016 Exchange semester in Master in ECONOMICS & PUBLIC POLICY
Universidad Adolfo Ibáñez – Chile

FIELDS OF INTEREST

Applied microeconomics (health, development, agriculture, climate change) & Food policy

ACADEMIC REFERENCES

Prof Franco Sassi	Imperial College London	f.sassi@imperial.ac.uk
Dr Laure de Preux	Imperial College London	l.depreux@imperial.ac.uk
Dr Olivier Ecker	International Food Policy Research Institute	o.ecker@cgiar.org

SCHOLARSHIPS AND AWARDS

2025 Imperial College London's Global Fellows Fund, **Turing Scheme**
Overseas research funding $\approx$ GBP 1,700

2021-2025 London Interdisciplinary Social Science Doctoral Training Partnership,
UK Economic and Social Research Council
Full-time tuition & stipend allowance + Research & training funds $\approx$ GBP 144,000

TEACHING EXPERIENCES

- FALL 2024 & 2025 Tutorial leader, MICROECONOMICS 2 (student feedback: [4.7/5 \(24\)](#); [4.6/5 \(25\)](#))
BSc Economics, Finance & Data Science; **Imperial College London**
Course leader: Prof. Tommaso Valletti (t.valletti@imperial.ac.uk)
- FALL 2024 Teaching Assistant, HEALTH ECONOMICS ([student feedback: 4.8/5](#))
MSc Global Health Management; **Imperial College London**
Course leader: Prof. Marisa Miraldo (m.miraldo@imperial.ac.uk)
- YEAR 2017-2018 Teacher, MATHEMATICS FOR ECONOMICS & FINANCE
Pre-BSc programme in Economics & Business; **University of Amsterdam**
Course leader: Evgenia Papoula (epapoula@oncampus.global)

PROFESSIONAL EXPERIENCES

- FEB 2025 - present Research assistant, Foresight & Policy Modeling Unit
International Food Policy Research Institute, Washington DC (remote)
- JUN 2023 - MAR 2024 Research assistant, Centre for Health Economics & Policy Innovation
Imperial College London, London, UK
- APR 2022 - JUL 2023 Consultant, Fiscal Policies for Health team
World Health Organization (WHO), Geneva, Switzerland (remote)
- MAR 2019 - AUG 2021 Consultant, Economics of Noncommunicable Diseases team
Risk Factors & Nutrition Unit
Pan American Health Organization (WHO - Americas), Washington DC
- DEC 2018 - MAR 2019 Consultant, Health, Nutrition & Population Global Practice,
World Bank, Washington DC

JOB MARKET PAPER

- [Smallholder Farming Households Nutrition Under Extreme Heat: Vulnerabilities and Adaptation](#), solo-authored
Abstract: Climate change threatens food security in Sub-Saharan Africa, but the magnitude of its nutritional impacts and the household-level behaviors that mediate them remain understudied. I examine how extreme heat during the crop growing season affects the commercialization and consumption decisions of Nigerian smallholder farming households, and how these responses shape post-harvest nutrition. Using five waves of panel data (2010–2024) linked to high-resolution weather, I exploit within-household variation in growing-season temperatures. Results show that hotter seasons depress harvests and reduce diet quality while leaving total caloric intake unchanged. I estimate that a +1°C warming would result in 1.62 and 0.63 million additional households with inadequate protein and iron intake, respectively. Less educated households and those with young children, who require protein and iron for growth, are more adversely affected. I demonstrate that households reduce crop commercialization to safeguard calories from own-produced staples and cut their purchases of nutritious and cash-intensive foods. This reflects a risk-minimizing behavior under incomplete food and labor markets and binding caloric constraints. In line with this, I find no evidence of adjustments in off-farm labor. Findings suggest that on-farm adaptation alone cannot maintain diet quality under climate change, underscoring the need for integrated policies that extend beyond calorie sufficiency and pair market participation and labor market development strategies with climate-risk mitigation.

PEER-REVIEWED ACADEMIC PUBLICATIONS

- [Taxation of foods high in fat, sugar, and sodium in India: A modelling study of health and economic impacts](#) (*PLOS Medicine*, 2026)
AUTHORS: **Maxime Roche (ICL)**, Jingmin Zhu (ICL), Jack Olney (ICL), Daniel J Laydon (ICL), William Joe (IEG), Manika Sharma (RTSL), Lindsay Steele (RTSL), Franco Sassi (ICL)
- [Can differentiated value-added tax rates promote healthier diets? The case of Costa Rica](#) (*Food Policy*, 2025), solo-authored
Media: [IB Knowledge blog](#)

WORKING PAPERS

- [The Future of Food Demand: Global Meta-Analysis and Projections of Income and Price Elasticities](#), *IFPRI Discussion Paper 2361*, Sep 2025; *under review*, with Andrew Comstock & Olivier Ecker (IFPRI)
Media: [IFPRI blog](#)
- [How is climate fuelling the thirst for sweetness? Exploring drivers and adaptation](#), *Chicago Booth Kilts Center for Marketing Data Center Paper Series*, May 2024, with Laure de Preux (ICL)

COMMENTARIES

- [The coming of age of food tax policy evaluations](#) (*The Lancet Public Health*, 2026)
AUTHORS: **Maxime Roche (ICL)**, Franco Sassi (ICL)

SELECTED PEER-REVIEWED POLICY PUBLICATIONS

- [Comparing taxes as a percentage of sugar-sweetened beverage prices in Latin America and the Caribbean](#) (*The Lancet Regional Health - Americas*, 2023)
AUTHORS: **Maxime Roche (ICL)**, Miriam Alvarado (Cambridge), Rosa Sandoval (PAHO), Fabio da Silva Gomes (PAHO), Guillermo Paraje (UAI)
- [Comparing taxes on alcoholic beverages in the Region of the Americas](#) (*Addiction*, 2023)
AUTHORS: **Maxime Roche (ICL)**, Rosa Sandoval (PAHO), Maristela Monteiro (PAHO)

POLICY BOOK CHAPTERS

- [United Nations. Handbook on Health Taxes for Developing Countries](#). Chapter 13: Specific Issues with respect to Taxation to Support Improved Nutrition (*accepted, forthcoming 2026*).
AUTHORS: Anne-Marie Thow (U Sidney), **Maxime Roche (ICL)**, Libby Hattersley (World Bank), Kate Mandeville (World Bank)
- [United Nations. Handbook on Health Taxes for Developing Countries](#). Chapter 9: Interaction between Health Taxes and other Instruments (*accepted, forthcoming 2026*).
AUTHORS: Franco Sassi (ICL), **Maxime Roche (ICL)**
- [World Health Organization. Health taxes: Policy and practice](#). Chapter 10: Monitoring and measuring health taxes (*World Scientific*, 2023)
AUTHORS: Rosa Sandoval (PAHO), **Maxime Roche (ICL)**, Anne-Marie Perucic (WHO), Itziar Belausteguigoitia (WHO), Miriam Alvarado (Cambridge), Luis Galicia (MoH URY), Guillermo Paraje (UAI)

ACADEMIC CONFERENCES & SEMINARS

2026 American Economic Association annual meeting (ASSA), University of York (*seminar*), Imperial College London (ICL)'s Public Health Policy Evaluation Unit (*seminar*)

2025 European Association of Agricultural Economists (EAAE) congress; Association of Environmental and Resource Economists (AERE) summer conference; International Food Policy Research Institute (IFPRI) (*seminar*); Potsdam Institute for Climate Impact Research DENeB PhD Workshop 'Climate Change, Agriculture, and Health'; Columbia University PhD Workshop in Sustainable

Development; University of Costa Rica's School of Economics (*seminar*); ICL's Economics & Public Policy (*seminar*); ICL's Centre for Health Economics & Policy Innovation (*seminar*)

2024 Agricultural and Applied Economics Association (AAEA); AERE; European Health Economics Association (EuHEA); Canadian Economics Association; French Economists Association; University of Goettingen's Sustainable Food Systems Symposium; Maastricht University's Workshop on Applied Economics of the Environment; ICL's Economics & Public Policy (*seminar*)

2023 IFPRI 'Delivering for Nutrition in South Asia' conference; University of Warwick 'Quantitative Methods in Social Sciences' workshop; University of Birmingham's Health Economics Unit (*seminar*); University of Costa Rica's School of Nutrition (*seminar*); ICL's Centre for Health Economics Policy Innovation (*seminar*); ICL's Public Health Policy Evaluation Unit (*seminar*)

SELECTED POLICY PRESENTATIONS

- **UK Department of Health & Social Care & World Health Organization, Regional Office for Europe**, Sugar and Calorie Reduction Network meeting: Fiscal levers (London, UK, 2025)
- **Pan American Health Organization**, Health tax policies in Latin America and the Caribbean: Are we making progress? (virtual, 2022, [link](#))
- **European Commission and ICF**, Mapping of fiscal measures and pricing policies applied to food, non-alcoholic and alcoholic beverages (virtual presentation, 2022, [link](#))
- **Congress of Paraguay**, Noncommunicable diseases: Scientific evidence for the use of fiscal policies for health (in Spanish) (Asuncion, Paraguay, 2019, [link](#))

INVITED REFEREE

JOURNALS	Health Economics; Food Policy; Applied Economics; The Lancet Public Health; PLOS One; Climatic Change
CONFERENCES	European Association of Agricultural Economists 2025; European Association of Young Economists 2025; European Health Economics Association 2024

OTHER PARTICIPATIONS & CONTRIBUTIONS

MEMBERSHIPS	American Economic Association; European Association of Agricultural Economists Agricultural & Applied Economics Association; Association of Environmental & Resource Economists; United Nations' Subcommittee on Health Taxes
COURSES	University of Copenhagen's Department of Food and Resource Economics 'International PhD course: Food and health policy analysis using home-scan data' (Copenhagen, Denmark, Dec 2023); BREAD-IGC 'Virtual PhD course on Environmental Economics' (virtual, 2023)
OTHER	Global Fellows Programme: 'Data for sustainability' (NTU Singapore, 2023); United Nations' Subcommittee on Health Taxes meeting (Geneva, 2025)

LANGUAGES

FRENCH: Mothertongue; ENGLISH: Fluent; SPANISH: Fluent

SOFTWARE SKILLS

Advanced Knowledge: STATA, Python, $\text{\LaTeX}$, Microsoft Office Suite
Working Knowledge: R, GIS, VBA, Tableau, GitHub